Analyzing Air Quality and Income Disparities in San Bernardino County, California

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Research question

How do PM2.5 levels compare with the percentage of low-income residents across census block groups in San Bernardino County?

Loading Packages

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
tmap_mode("plot")
```

Reading the EJScreen data

```
ejscreen <- sf::st_read( #Read in EJScreen data
  here::here(
    "data",
    "ejscreen",
    "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
  ),
  quiet = TRUE # Used for pdf
)
```

Filtering to California

```
california <- ejscreen |>
  dplyr::filter(ST_ABBREV == "CA")
```

Filtering the data to San Bernardino County in California

```
san_bernardino_bg <- california |>
  dplyr::filter(CNTY_NAME %in% c("San Bernardino County"))
nrow(san_bernardino_bg)
```

```
[1] 1275
```

Check air quality and income values

```
san_bernardino_bg |>
  sf::st_drop_geometry() |>
  dplyr::select(ID, PM25, LOWINCPCT) |>
  summary()
```

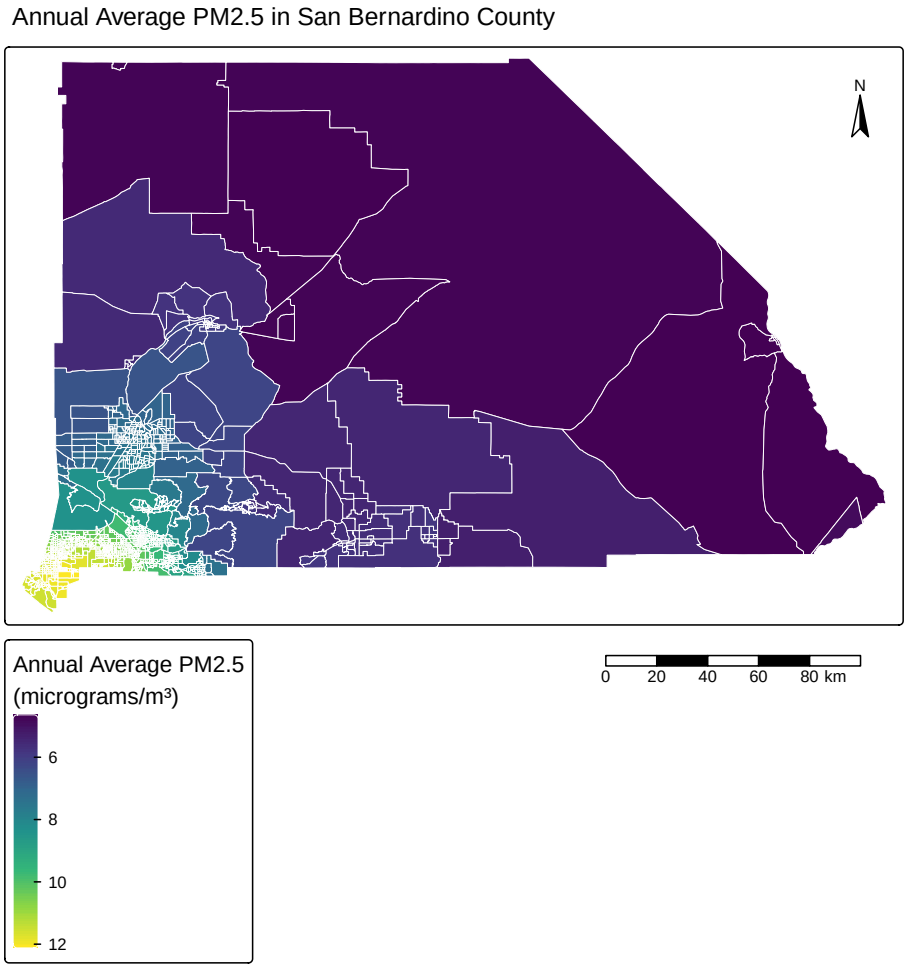
```
      ID                 PM25           LOWINCPCT      
 Length:1275        Min.   : 4.608   Min.   :0.0000  
 Class :character   1st Qu.: 7.643   1st Qu.:0.1890  
 Mode  :character   Median : 9.967   Median :0.3260  
                    Mean   : 9.437   Mean   :0.3479  
                    3rd Qu.:10.978   3rd Qu.:0.4880  
                    Max.   :12.137   Max.   :1.0000  
```

Map 1, looking at PM 2.5 Levels in San Bernardino County, California

```
# Map 1: Annual-average PM2.5 concentrations
pm25_map <- tm_shape(san_bernardino_bg) +
  tm_polygons(
    fill = "PM25",
```

```
    lwd = 0.1, #creating borders
    col = "white",# setting border color
    fill.scale = tm_scale_continuous( #continuous color scale to see different concentration
      values = "viridis"
    ),
    fill.legend = tm_legend( #creating legend
      title = "Annual Average PM2.5\n(micrograms/m³)",
      orientation = "portrait",
      position = tm_pos_out(
        "center", "bottom",
        pos.h = "left", # legend alignment
        pos.v = "top"
      )
    )
  ) +
  tm_compass( #add North arrow
    size = 2,
    text.size = 0.7,
    position = tm_pos_in("right", "top") # put it in upper right corner
  ) +
  tm_scalebar( # add scalebar
    width = 15,
    text.size = 0.7,
    position = tm_pos_out(
      "center", "bottom",
      pos.h = "right", # horizontal position
      pos.v = "top" # vertical position
    )
  ) +
  tm_title( # add title
    "Annual Average PM2.5 in San Bernardino County",
    size = 1,
    position = tm_pos_out(
      "center", "top", # title above map
      pos.h = "left",# horizontal position
      pos.v = "bottom"# vertical position
    )
  )

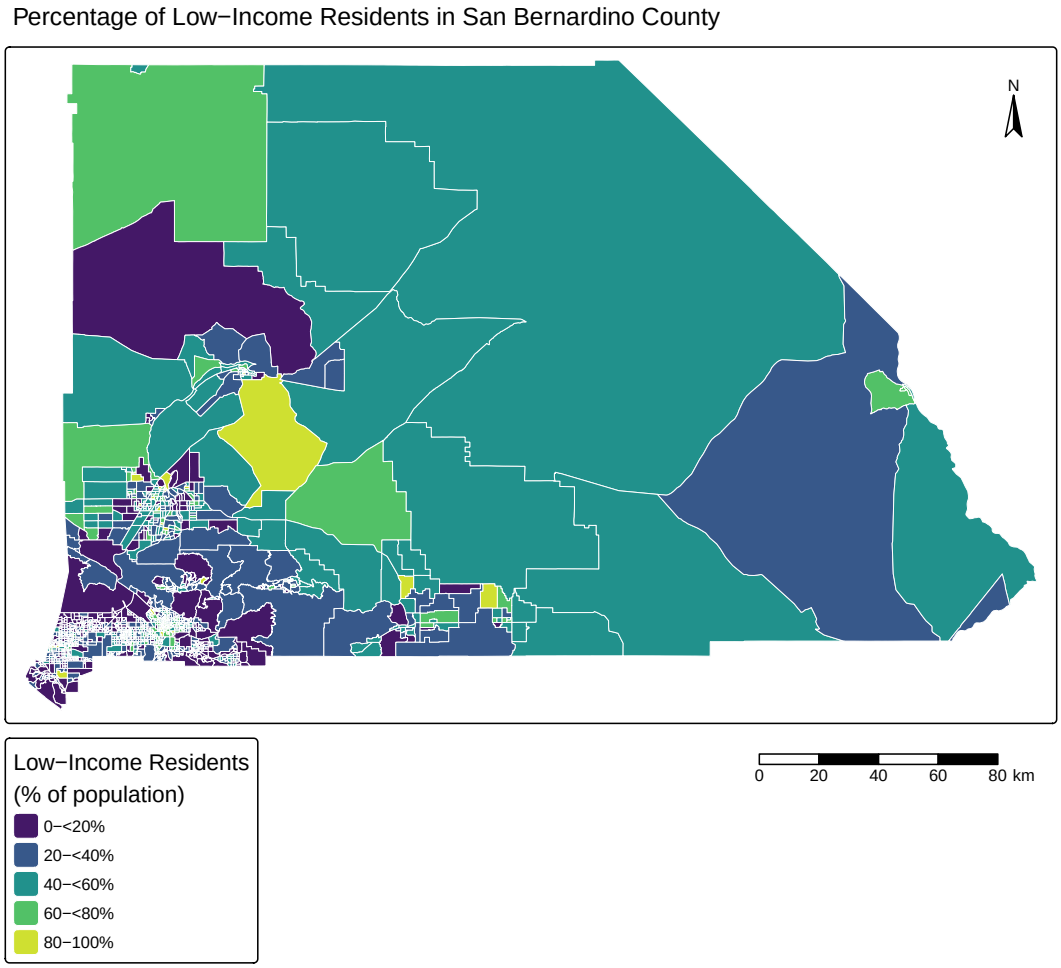
pm25_map # Display map
```



Map 2: Percentage of low-income residents

```
income_map <- tm_shape(san_bernardino_bg) +
  tm_polygons(
    fill = "LOWINCPCT",
    lwd = 0.1, # borders
    col = "white", # borders
    fill.scale = tm_scale_intervals( #creating intervals for more readable map
      style = "fixed",
      breaks = c(0, 0.2, 0.4, 0.6, 0.8, 1), # 5 categories
```

```
    labels = c(
      "0-<20%",
      "20-<40%",
      "40-<60%",
      "60-<80%",
      "80-100%"),
    values = "viridis"),
  fill.legend = tm_legend( #map legend
    title = "Low-Income Residents\n(% of population)",
    orientation = "portrait",
    position = tm_pos_out(
      "center", "bottom", # legend alignment
      pos.h = "left", # horitzontal position
      pos.v = "top" )) ) + # vertical position
tm_compass( # north arrow placement
  size = 2, text.size = 0.7,
  position = tm_pos_in("right", "top")) +
tm_scalebar( #scalebar and its placement
  width = 15,
  text.size = 0.7,
  position = tm_pos_out(
    "center", "bottom",
    pos.h = "right",
    pos.v = "top")) +
tm_title( "Percentage of Low-Income Residents in San Bernardino County",
  size = 1,
  position = tm_pos_out(
    "center", "top",
    pos.h = "left", #horizontal position
    pos.v = "bottom"  # veritcal position
  ))
  income_map # display map
```



Map Results

The first map shows annual-average PM2.5 concentrations across census block groups in San Bernardino County. Concentrations are higher in the southwestern area of the county, while the northern and eastern areas have lower concentrations. The second map shows the percentage of lowm income residents in the county, and its pattern differs from the first map. In the southwestern area, there is a mix of block groups with both low and high percentages of low income residents. Some block groups farther north have higher percentages of low income residents and lower PM2.5 concentrations. Together, the maps suggest that higher PM2.5 concentrations do not consistently overlap with higher percentages of low-income residents across the county. It may be beneficial to examine the southwestern region more closely to

determine whether block groups with higher percentages of low-income residents also tend to have higher PM2.5 concentrations within that condensed region.